

and connectivity requirements of the device.

The cloud provides additional storage along with local onboard storage to achieve anytime access, data security, and unlimited extendible storage. The user application layer is key to the success of the product as customers get a feel using it. A user can be a chief technology officer (CTO), analysing a dashboard to understand average defect rate, machine breakdowns, and production numbers, or a software engineer at the office checking what's the average temperature at home to switch on the AC or geyser using a mobile app.

Data analytics

After IoT data analysis, businesses can develop machine learning (ML) models and implement them as part of the IoT solution infrastructure, so that action is taken without any human intervention when a similar pattern arises in the data generated in real time. ML trains machines on how to respond to a combination of inputs, even unknown inputs, and ensure a steady and accurate output. It involves performing data modelling on the data collected by IoT sensors, followed by creating an ML model using statistical or mathematical algorithms. These models are deployed and used until they ensure high accuracy.

In healthcare, the IoT revolution is driving new equipment, remote monitoring, and wearables to constantly monitor the health of individuals. Performing data analytics can help draw useful insights from the data like heart rate pattern, SO_2 oxygen pattern, and average body temperatures. Based on these vitals, doctors can predict symptoms of disease well in advance.

"You can have data without information, but you cannot have information without data," says Daniel Keys Moran, an American fiction writer and computer programmer. IoT is key to look deep into processes and generate related data, whereas



Figure 2: The IoT technology stack contains five layers to provide all the capabilities of an IoT solution (Credit: <https://iotforall.com>)

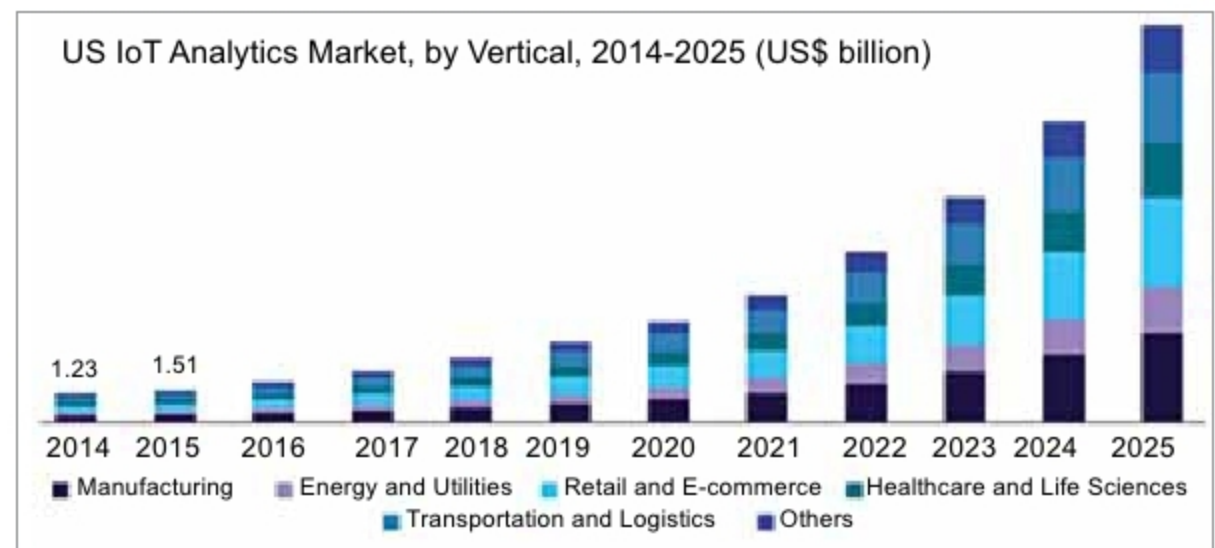


Figure 3: The global (IoT) analytics market size is valued at US\$ 5.7 billion. It is expected to progress at a CAGR of 29.7 per cent over the forecast period (Credit: www.grandviewresearch.com)

analytics is a must to dive deep into data to produce useful insights. IoT and analytics remain connected and grow together.

IoT analytics

IoT analytics emerged for crunching the IoT data. It refers to data tools used to draw values from Big Data generated by the IoT. A lot of companies implement IoT technology to gather data from sensors placed on solar panels, delivery trucks, machinery, and human bodies. A lot of related data comes from gateways, data storage, and network devices. IoT data is streamed at high volumes and high data rates continuously and considered a special case of Big Data.

These streams of data are combined and transformed to present consumable reports. IoT's real-time processing and insights help companies make better marketing and operational decisions. IoT data poses challenges, including data integration, compatibility, and data analytics, which are well addressed using IoT analytics.

Analysts estimate the IoT analytics market to grow at a CAGR of 29.7 per cent by 2025, at a valuation of 5.7 billion dollars. While smart cities, smartwatches, and other 'smart' initiatives are driving IoT adaption, social media, mobile, Internet, and IoT itself are driving the IoT analytics. **END** 🐧

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